Spectral Simulation Report

Automated Analysis

June 18, 2025

Contents

1 Introduction **2**

2 Sample: 11041 **2**
2.1 Simulation Parameters 2
2.2 Composition Analysis 2
2.3 Spectral Plot 2

3 Sample: 7548 **4**
3.1 Simulation Parameters 4
3.2 Composition Analysis 4
3.3 Spectral Plot 4

4 Sample: 6931 **6**
4.1 Simulation Parameters 6
4.2 Composition Analysis 6
4.3 Spectral Plot 6

5 Sample: 14436 **8**
5.1 Simulation Parameters 8
5.2 Composition Analysis 8
5.3 Spectral Plot 8

6 Sample: 5892 **10**
6.1 Simulation Parameters 10
6.2 Composition Analysis 10
6.3 Spectral Plot 10

1 Introduction

This report presents the results of spectral simulations derived from HDF5 dataset analysis. Each sample includes a spectral plot and a detailed table summarizing the initial composition and actual labeled positions of elements, along with simulation parameters such as temperature and electron density.

2 Sample: 11041

2.1 Simulation Parameters

Parameter	**Value**
Temperature (T)	6919 K
Electron Density (n_e)	7.39e+16 $\mathrm{cm}^{-}3$

Table 1: Simulation parameters for sample 11041.

2.2 Composition Analysis

Element	**Initial Composition (%)**	**Actual Labeled Position (%)**
Al	0.00	0.00
Ar	10.13	9.59
C	7.65	5.32
Ca	9.44	3.66
Cl	3.82	0.29
Co	16.13	25.71
Cr	12.66	9.69
Fe	0.00	0.00
Mg	0.00	0.00
Mn	0.00	0.00
N	0.00	0.00
Na	7.21	8.06
Ni	3.84	11.67
O	0.00	0.00
S	13.52	0.61
Si	15.61	4.93
Ti	0.00	0.00
background	N/A	46.00
Total (%)		125.54

Table 2: Composition analysis for sample 11041.

2.3 Spectral Plot

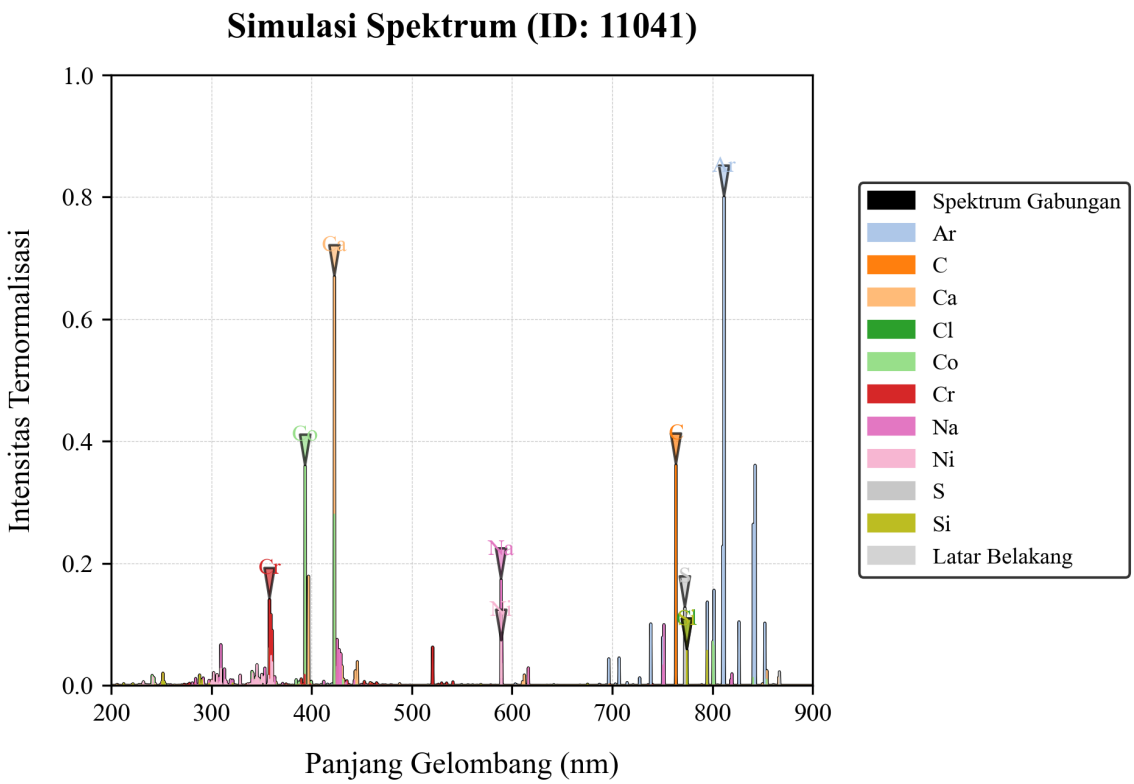

Figure 1: Spectral simulation plot for sample 11041.

3 Sample: 7548

3.1 Simulation Parameters

Parameter	Value
Temperature (T)	8838 K
Electron Density (n_e)	3.13e+15 $\mathrm{cm}^{-}3$

Table 3: Simulation parameters for sample 7548.

3.2 Composition Analysis

Element	Initial Composition (%)	Actual Labeled Position (%)
Al	0.00	0.00
Ar	0.00	0.00
C	0.00	0.00
Ca	20.96	3.66
Cl	20.48	0.76
Co	9.75	34.28
Cr	21.92	9.69
Fe	0.00	0.00
Mg	10.79	8.47
Mn	1.61	17.24
N	0.00	0.00
Na	6.62	7.86
Ni	5.59	11.67
O	0.00	0.00
S	1.77	4.37
Si	0.50	5.08
Ti	0.00	0.00
background	N/A	38.28
Total (%)		141.36

Table 4: Composition analysis for sample 7548.

3.3 Spectral Plot

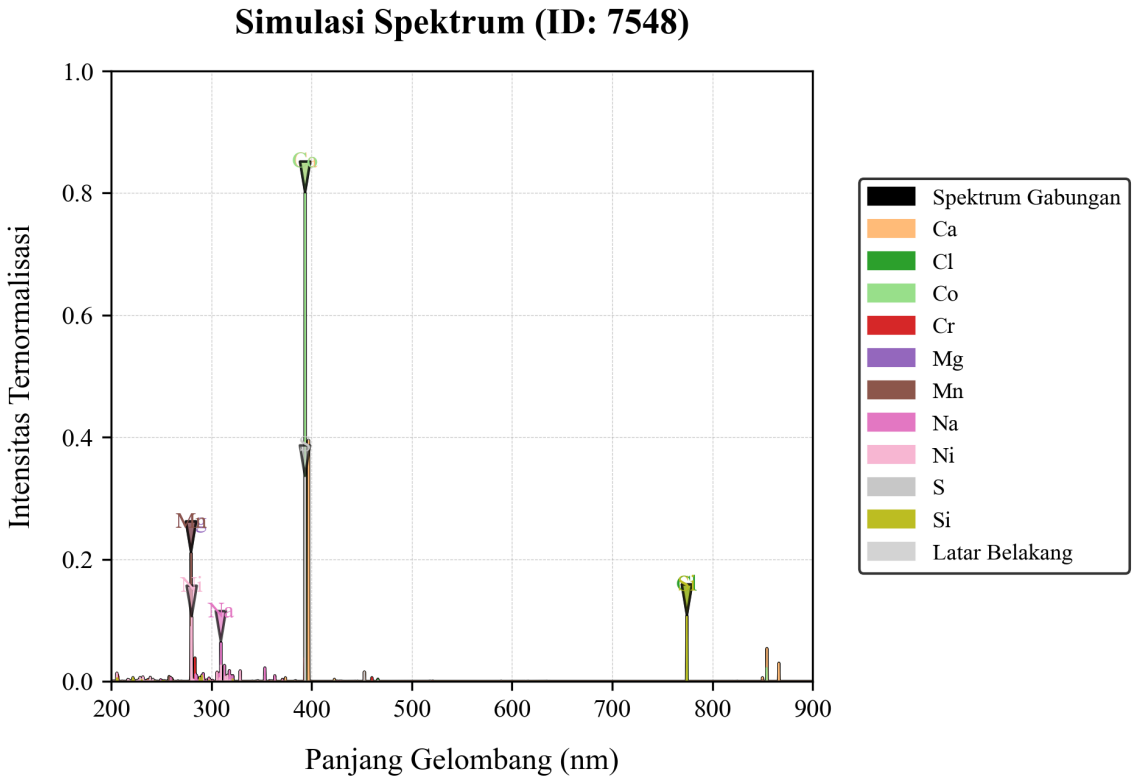

Figure 2: Spectral simulation plot for sample 7548.

4 Sample: 6931

4.1 Simulation Parameters

Parameter	Value
Temperature (T)	9747 K
Electron Density (n_e)	1.48e+15 $\mathrm{cm}^{-}3$

Table 5: Simulation parameters for sample 6931.

4.2 Composition Analysis

Element	Initial Composition (%)	Actual Labeled Position (%)
Al	0.00	0.00
Ar	0.00	0.00
C	0.00	0.00
Ca	12.47	3.66
Cl	3.08	0.81
Co	0.00	0.00
Cr	2.53	9.69
Fe	0.00	0.00
Mg	6.70	8.47
Mn	13.08	17.24
N	6.12	9.79
Na	7.47	7.86
Ni	11.91	11.67
O	7.55	4.47
S	10.08	5.57
Si	10.74	5.44
Ti	8.26	28.96
background	N/A	35.42
Total (%)		149.05

Table 6: Composition analysis for sample 6931.

4.3 Spectral Plot

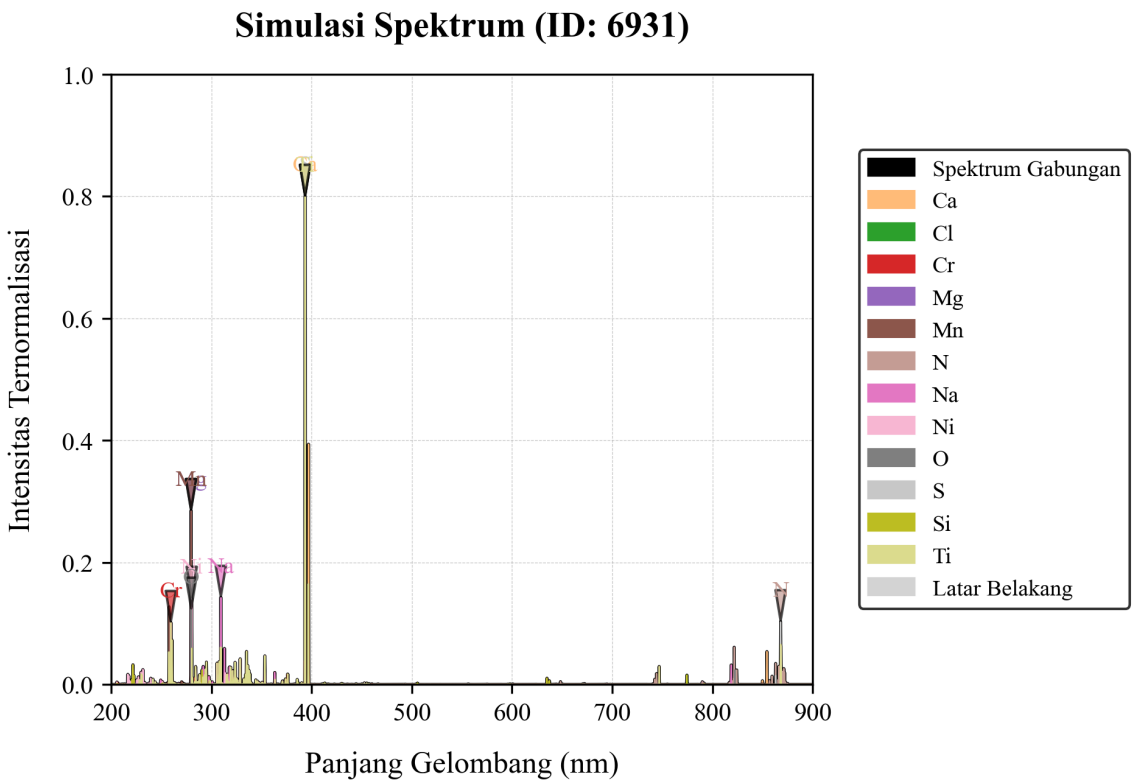

Figure 3: Spectral simulation plot for sample 6931.

5 Sample: 14436

5.1 Simulation Parameters

Parameter	Value
Temperature (T)	11061 K
Electron Density (n_e)	6.58e+15 $\mathrm{cm}^{-}3$

Table 7: Simulation parameters for sample 14436.

5.2 Composition Analysis

Element	Initial Composition (%)	Actual Labeled Position (%)
Al	4.04	11.28
Ar	10.71	17.75
C	0.00	0.00
Ca	9.80	3.66
Cl	11.62	0.85
Co	0.00	0.00
Cr	4.15	9.69
Fe	0.00	0.00
Mg	5.31	8.59
Mn	7.98	17.24
N	0.77	9.89
Na	7.19	7.86
Ni	12.80	11.67
O	7.07	4.79
S	13.90	5.57
Si	4.66	5.44
Ti	0.00	0.00
background	N/A	32.08
Total (%)		146.36

Table 8: Composition analysis for sample 14436.

5.3 Spectral Plot

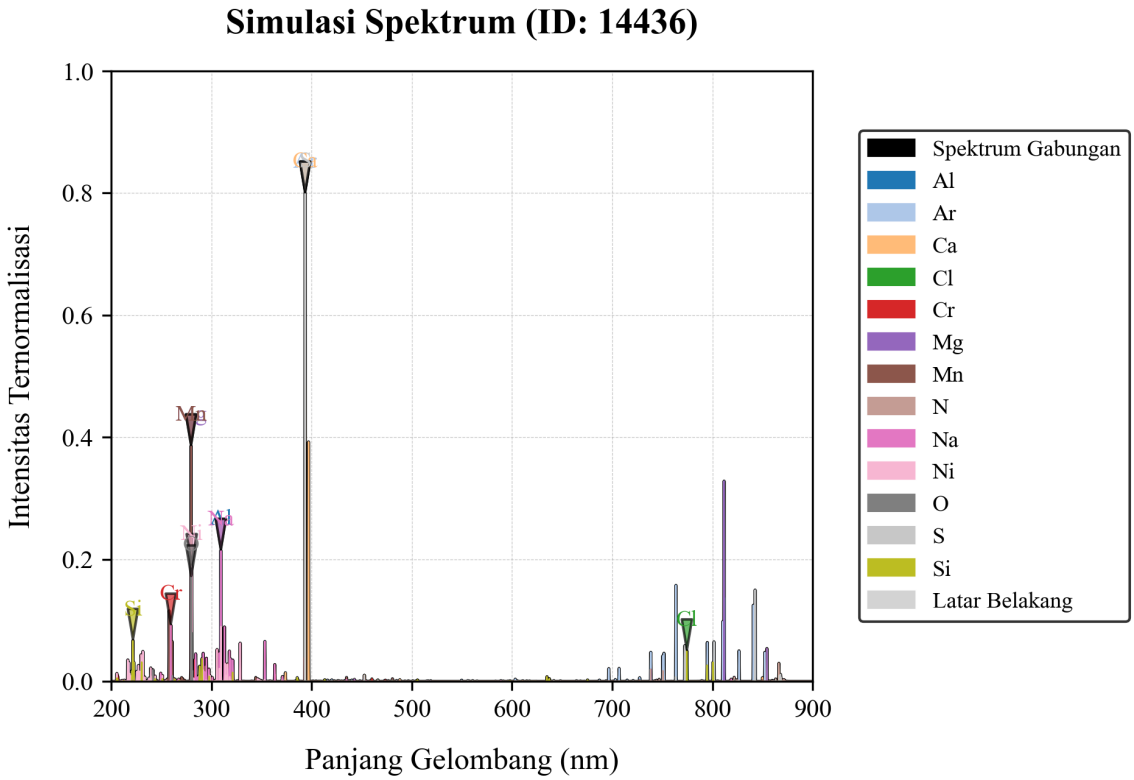

Figure 4: Spectral simulation plot for sample 14436.

6 Sample: 5892

6.1 Simulation Parameters

Parameter	Value
Temperature (T)	12778 K
Electron Density (n_e)	9.55e+15 $\mathrm{cm}^{-}3$

Table 9: Simulation parameters for sample 5892.

6.2 Composition Analysis

Element	Initial Composition (%)	Actual Labeled Position (%)
Al	0.00	0.00
Ar	0.00	0.00
C	0.00	0.00
Ca	1.54	3.66
Cl	14.47	0.85
Co	0.00	0.00
Cr	14.15	9.69
Fe	7.44	67.41
Mg	10.37	8.64
Mn	6.01	17.24
N	7.83	16.14
Na	5.92	7.86
Ni	15.26	11.67
O	0.00	0.00
S	7.46	5.57
Si	9.55	5.44
Ti	0.00	0.00
background	N/A	18.92
Total (%)		173.10

Table 10: Composition analysis for sample 5892.

6.3 Spectral Plot

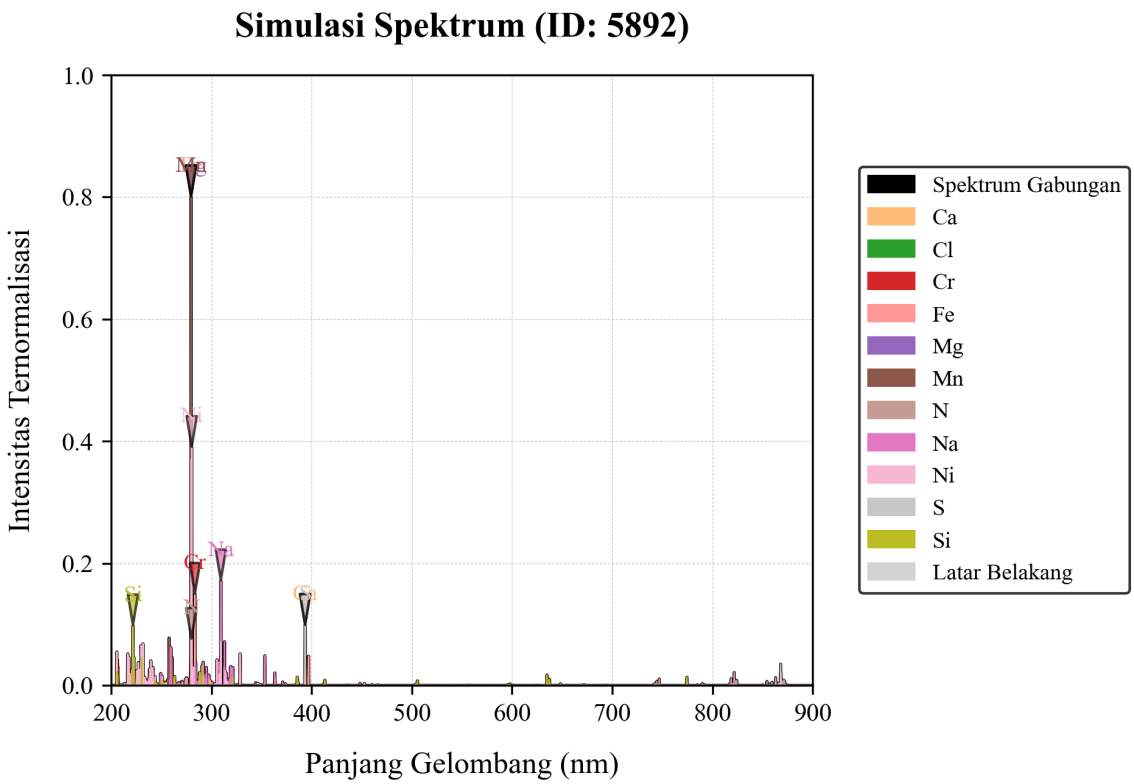

Figure 5: Spectral simulation plot for sample 5892.